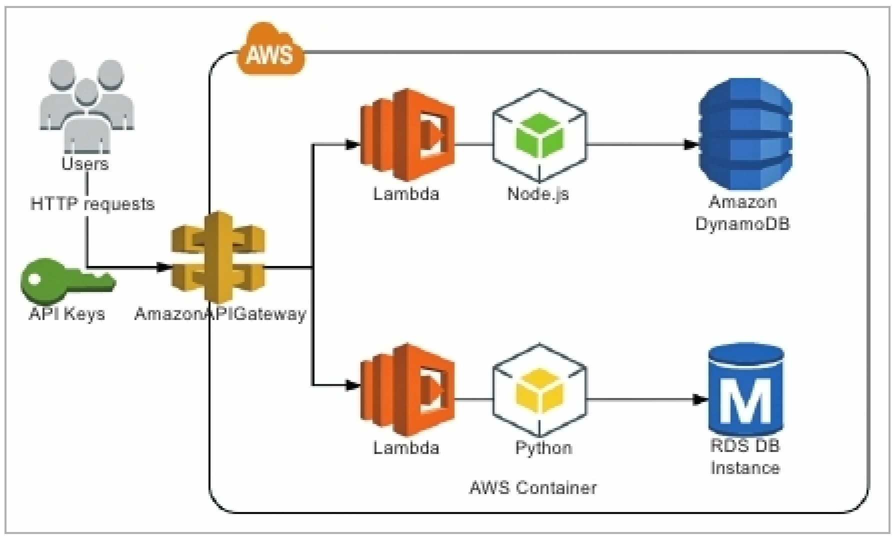

Figure 6: AWS API gateway use case for serverless application deployments

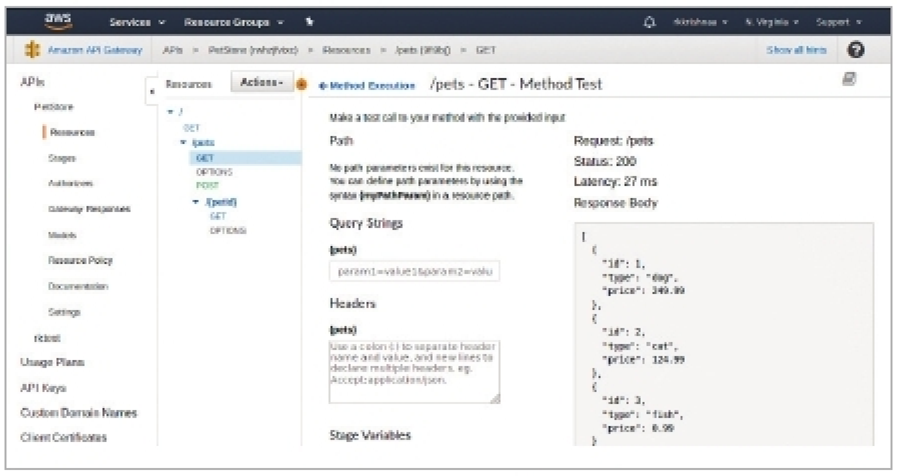

Figure 7: Management console to manage the AWS API gateway

A custom domain can be created and SSL certificates used to access API endpoints. API keys can be created and imported to access API endpoints. You can also create VPC links from the API gateway to network load balancers. The performance of the API can be tracked easily with the help of Cloudwatch dashboards.

APIs can be easily documented from the management console. New APIs can be created through a a Swagger file. It is very easy to export or import REST APIs from one environment to another. You can generate client SDKs for the API to support platforms like Java, Android, iOS, JavaScript and Ruby.

API deployments can be facilitated by writing SAM templates. Here, you can define the stages and other dependencies as Swagger files.

An introduction to Swagger

Swagger offers the most powerful and easy-to-use tools to take full advantage of the OpenAPI Specification. OpenAPI Specification (formerly Swagger Specification) is an API description format for REST APIs. An OpenAPI file allows us to describe an entire API, including the available endpoints (*/users*) and operations on each endpoint (*GET / users*, *POST /users*), operation parameter inputs and outputs, authentication methods, contact information, licences, terms of use and other information.

API specifications can be written in YAML or JSON. The format is easy to learn and readable for both humans and machines. Swagger is a powerful yet easy-to-use suite

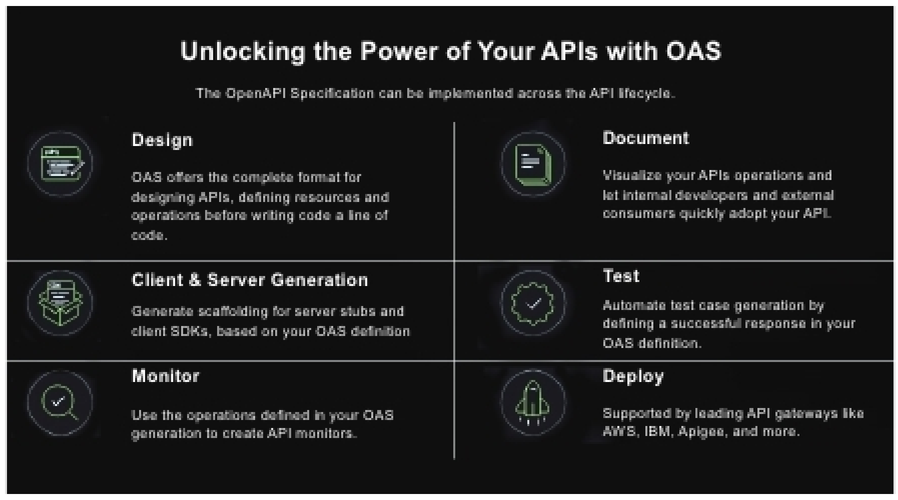

Figure 8: Why Swagger?

of API developer tools for teams and individuals, enabling development across the entire API lifecycle -- from design and documentation, to test and deployment. It consists of a mix of open source, free and commercially available tools that allow anyone, from technical engineers to streetsmart product managers, to build amazing APIs that everyone loves.

The major Swagger tools include: Swagger Editor, Swagger UI and Swagger Codegen. Each tool can be used for different purposes. Swagger Editor allows you to write a spec for REST APIs and Swagger UI helps to create API documentation based on the Open API Specification. Swagger Codegen helps to generate client libraries in different programming languages. Swagger is available in both open source as well as proprietary versions.

The best example to get started with writing a Swagger schema for REST APIs, which can be expressed in the JSON or YAML format, is given at *https://petstore.swagger.io/*. This example implements the complete documentation of a pet store. This JSON definition covers all the information related to the API such as the version, description, title, licence, host, port, URL resources, scheme, request parameters, request data, response parameters, response data, response code and additional messages. A mock environment can be created to test the RESTful APIs.

The example given below explains how to create a simple movie store API with Swagger.

```
swagger: "2.0"          # Swagger version

info:                   # This object provides metadata
about the API.
  version: 1.0.0
  title: Movie Store
  description: A simple Movie Store API

basePath: /api          # Path relative to which all API
calls will be made

paths: {}               # This Object defines the paths/
endpoints of your API
```